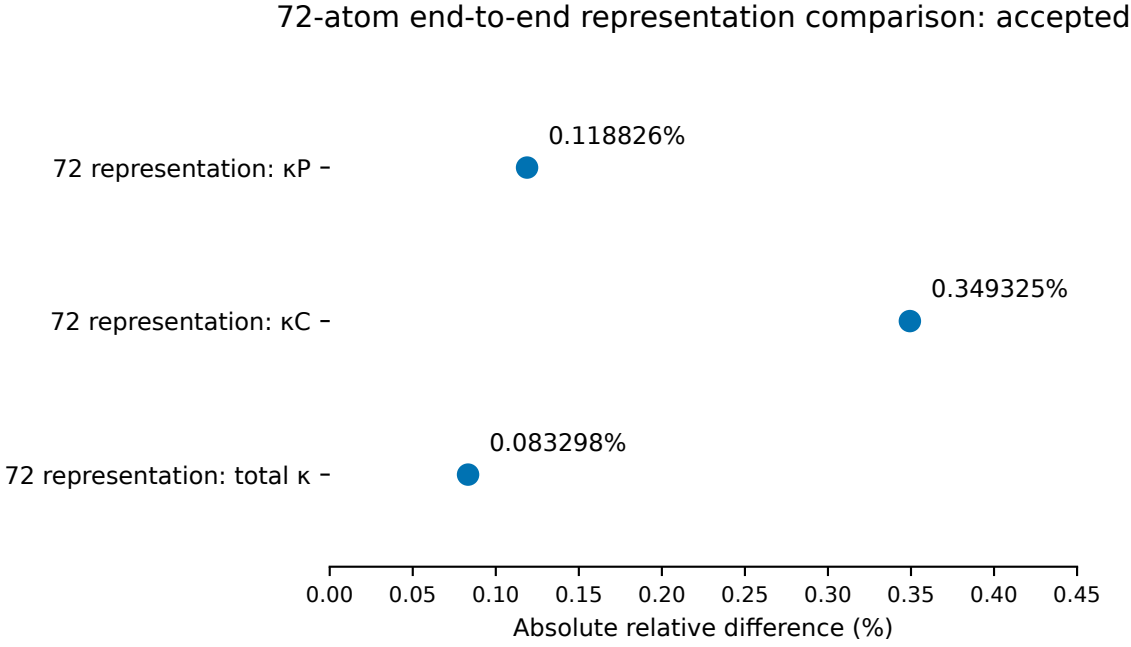

73-atom modal κP / κC: max tolerance ratios 0.006343 / 8.811e-6 (PASS).

All 24 q points × 219 branches: max |Δγ| = 1.961e-14 THz vs streamed NumPy.